

Azure Cloud Honeypot & SIEM Threat Detection Lab

A deliberately exposed Windows VM in Azure, wired into Microsoft Sentinel to capture, enrich, and visualize real brute-force login attempts from across the internet.

PLATFORM	SIEM	QUERY LANGUAGE	DATA VOLUME
Microsoft Azure	Microsoft Sentinel	KQL	~54,000-row GeoIP watchlist

Log Analytics Workspace — Logs

```
// enrich failed RDP logons with attacker geolocation
let GeoIPDB_FULL = _GetWatchlist("geoip");
SecurityEvent
| where EventID == 4625
| order by TimeGenerated desc
| evaluate ipv4_lookup(GeoIPDB_FULL, IPAddress, network)
```

TIMEGENERATED	ACCOUNT	IPADDRESS	CITY	COUNTRY	RESULT
2026-08-11 03:14:02	employee	185.220.xx.xx	—	RU	4625
2026-08-11 03:11:47	administrator	45.146.xx.xx	—	NL	4625
2026-08-11 02:58:19	test	103.99.xx.xx	—	CN	4625

OVERVIEW

Why build a honeypot

Exposing a VM to the open internet on purpose turns it into a sensor. Every scan, credential-stuffing attempt, and brute-force login becomes a labeled data point — the same signal a SOC analyst triages in production, but generated on demand instead of waiting for it to happen. This project builds the full detection pipeline behind that signal: exposure, capture, forwarding, enrichment, and visualization.

4625

Event ID monitored (failed logon)

54K

IP ranges in GeoIP watchlist

6

Pipeline stages, end to end

1

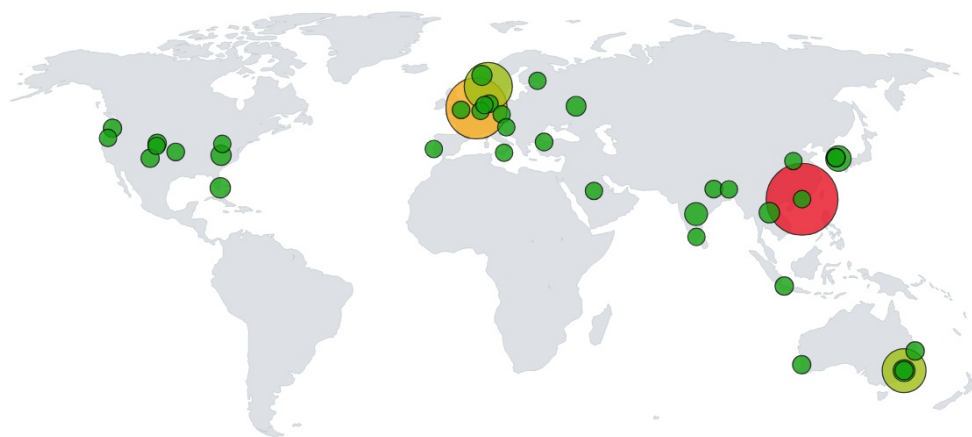
Live geographic attack map

LIVE RESULT

Global attack map

Rendered in a Sentinel Workbook using the GeoIP-enriched watchlist join — every bubble is a real inbound RDP attack against the honeypot, sized by attempt volume.

Sentinel Workbook — Windows VM Attack Map



-(Hong Kong)	Maam (Netherlands)	Skien (Norway)	Sydney (Australia)	Other	Jung-gu (South Korea)	Hyderabad (India)	Sydney (Australia)	Nong Khai (Thailand)	Hollywood (United States)
63.1 k	45.2 k	27 k	21.9 k	9.43 k	4.83 k	3.15 k	2.52 k	1.69 k	1.6 k

63.1K

Top source — Hong Kong

45.2K

Netherlands

27K

Norway

10+

Countries observed attacking

ARCHITECTURE

Pipeline

Six stages, from exposed endpoint to a queryable, mapped view of who's knocking.

01

Expose the target

Windows VM, inbound NSG rule opened, host firewall disabled.

02

Generate signal

Failed RDP logons captured locally as Event ID 4625.

03

Forward logs

AMA agent + DCR ship Security events to a Log Analytics Workspace.

04

Centralize in SIEM

Sentinel connected to the workspace for query and detection.

05

Enrich with geo data

54K-row GeoIP CSV imported as a Sentinel Watchlist, joined via KQL.

06

Visualize

Custom Sentinel Workbook renders attacker origins on a live map.

SKILLS DEMONSTRATED

What this proves

[CLOUD]

Cloud infrastructure

Provisioned and network-configured a VM in Azure, including NSG rules and security posture trade-offs.

[SIEM]

SIEM operation

Connected and configured Microsoft Sentinel against a live Log Analytics Workspace.

[QUERY]

KQL log analysis

Wrote Kusto queries to filter, join, and enrich large-scale security event data.

[INTEL]

Threat intelligence

Applied a 54,000-row GeoIP dataset to convert raw IPs into attacker geolocation.

[VIZ]

Dashboarding

Built a custom JSON-configured Sentinel Workbook to map attacks geographically.

[OPS]

Cost-aware operations

Managed compute spend against a fixed budget by right-sizing and deallocating on idle.

Microsoft Azure

Microsoft Sentinel

Log Analytics Workspace

KQL

Azure Monitor Agent

Data Collection Rules

NSG / Network Security

Threat Intelligence Enrichment

SOC Fundamentals

Built as a hands-on SOC / detection engineering lab.

[View the write-up on GitHub →](#)